



Luca Sportelli

DOCTORAL RESEARCHER
SPACECRAFT AUTONOMY

CONTACT

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PROFILES

Google Scholar
GitHub
LinkedIn

RESEARCH FOCUS

AI for GNC
Rendezvous and proximity operations
Vision-based navigation
Small-body navigation
Interpretable autonomy

CORE TOOLS

Python
C/C++
MATLAB/Simulink
STK
ESATAN
L^AT_EX

Academic CV

01 / 02

RESEARCH / ENGINEERING / EXPLORATION

AI for spacecraft GNC and autonomy

Doctoral researcher developing interpretable AI for spacecraft guidance, navigation and control. Research focuses on temporal vision-based navigation, autonomous rendezvous and proximity operations, AI-augmented model predictive control, and autonomous planetary systems. Earlier work spans small-body imaging, spacecraft thermal engineering, mission design and analog astronaut operations.

RESEARCH PAPERS & THESIS

AI-Augmented Model Predictive Control for Safe and Adaptive Rendezvous and Proximity Operations

L. Sportelli, T. Barr, C. Kilic, D. Wu · arXiv preprint · 2026

Design Principles for Explainable and Adaptive AI in Space Differential Games

T. Barr et al. (incl. L. Sportelli) · AAS/AIAA Astrodynamics · 2026

Modeling Particle Propagation in the Didymos Binary System for Imaging Simulation and Particle Rendering

L. Sportelli · MSc thesis · Politecnico di Milano · 2024

Structural and Thermal Model Testing Campaign of a 1U CubeSat

F. Botti et al. (incl. L. Sportelli) · 75th IAC · 2024

Design and Architecture of Anubis: A Hopper to Sample and Study Mercury's Surface and Subsurface

A. Cagnoni et al. (incl. L. Sportelli) · 75th IAC · 2024

A Student Approach for Thermal Modelling, Validation and Testing of the 6S CubeSat

D. Scalettari et al. (incl. L. Sportelli) · 74th IAC · 2023

RESEARCH & ENGINEERING EXPERIENCE

Doctoral Researcher · XDLab, ERAU

Aug 2025–Present

Developing AI-enabled methods for autonomous spacecraft navigation and control, with current research spanning temporal vision-based state estimation for rendezvous, proximity operations and small-body navigation, as well as AI-augmented model predictive control for safe and adaptive spacecraft operations. Research emphasizes interpretable AI, physics-informed learning and robust integration with classical GNC methods. Advisor: Prof. Di Wu.

Thermal Control System Team Leader · PoliSpace 6S

Apr 2022–Mar 2024

Led thermal design and analysis for the student-developed 1U CubeSat, including thermal strategy definition and ESATAN and MATLAB/Simulink modelling. As Team Leader, responsible for team management and system interfaces. Team selected for ESA's Fly Your Satellite! Design Booster programme; work presented at the International Astronautical Congress in Baku and Milan.

EDUCATION & AWARDS

PhD Aerospace Eng. · Embry-Riddle Aeronautical University

Aug 2025–Present

Research on AI-enabled spacecraft autonomy, navigation and control. Advisor: Prof. Di Wu.

MSc Space Engineering · Politecnico di Milano

Sep 2022–Dec 2024

110/110 cum laude. Thesis on particle propagation and imaging simulation in the Didymos binary system. Fabrizio Schiavi scholarship; DSU scholarships in 2022 and 2023.

Alta Scuola Politecnica · Politecnico di Milano

Feb 2023–Aug 2024

Highly selective multidisciplinary programme pursued alongside the MSc, combining additional coursework with team-based interdisciplinary projects. Served as Team Controller for the programme's project team. **Selected coursework:** Dynamics of Innovation; Design Methods; The Intangible Matters of Places: Impact and Innovation Commons; Management for Sustainability and Impact; High-Tech Startups.

BSc Aerospace Engineering · Politecnico di Milano

Sep 2019–Jul 2022

108/110. DSU scholarships in 2019, 2020 and 2021.

Clickable links provide access to profiles, publications, reports and project documentation.

LS

Luca Sportelli

Experience / technical work

EXTENDED TECHNICAL TOOLKIT

Analysis and simulation

Simscape

Dymola

GMAT

NASA CEA

Systems and design

DOORS

Capella / MBSE

Yamcs Mission Control

Solid Edge

SolidWorks

Autodesk Inventor

CERTIFICATIONS

IELTS Academic 8.0

L 9.0 · R 8.5 · W 6.5 · S 7.0

IPMA Introductory Certificate

GRE, Mar 2022

Ancient Latin Language Certification, May 2017

LANGUAGES

English C1

French A1

Russian A1

Ancient Latin A1

ADDITIONAL

Valid driving licence

Soccer player · ASD RGP Precotto

Trained hip-hop dancer

Blood donor · Policlinico di Milano

Academic CV

02 / 02

FIELD WORK / SYSTEMS / MISSION DESIGN

Experience and selected projects

ANALOG MISSIONS & OTHER EXPERIENCE

Analog Astronaut · Asclepios IV, Selene Crew

Aug 2023–Oct 2024

Selected worldwide as one of eight Selene Crew analog astronauts through a six-month ESA-like selection process, followed by one year of mission training. Training included extreme-environment operations, firefighting, zero-gravity flight, interview and communication techniques, CapCom activities in mission control, virtual-reality lunar landing simulations and experiment preparation. The programme was supported by mentors including former ESA astronaut Claude Nicollier and former ESA expert Prof. Marc Toussaint.

Analog Astronaut / Data Officer · EuroMoonMars EMMPOL 13

Oct 2022

Participated in a two-week international analog mission in Poland with a six-person, six-nationality crew under isolation, no-sunlight and sleep-deprivation conditions, with emphasis on emergency procedures and individual research activities. Served as Data Officer and conducted a study on the thermal interaction between a human occupant and a confined habitat environment.

MEEVA Project Member · PoliSpace & Asclepios

Oct 2022–Apr 2023

Designed and integrated the cooling system for an analog astronaut suit as part of the Medical Equipment for EVA project.

Peer Tutor · Politecnico di Milano

2020–2022

Academic tutoring during the 2020–21 and 2021–22 academic years.

SELECTED ACADEMIC PROJECTS

MIRAGE Mercury Mission

Sep 2023–Feb 2024

Developed a twelve-student Mercury mission architecture comprising an interplanetary transfer module, orbiter, science orbiter and hopper. The mission targeted volatile-rich terrain through three-site surface sampling and investigated Mercury's mineralogy, interior structure and dynamic exosphere.

MESSENGER Reverse Engineering

Feb–Jul 2023

Performed a reverse-engineering study of NASA's MESSENGER spacecraft to assess how mission constraints shaped system architecture, subsystem sizing and major design decisions.

6U CubeSat ADCS

Oct 2022–Jan 2023

Designed detumbling and Earth-pointing attitude determination and control for a 6U CubeSat in LEO, including spacecraft dynamics and closed-loop simulation in MATLAB/Simulink.

Interplanetary & Planetary Explorer

Oct 2022–Jan 2023

Performed a feasibility study for an interplanetary mission to asteroid VF32 Aten with an intermediate Jupiter flyby. The project also investigated the effects of Earth's J2 perturbation and lunar gravity on Keplerian orbital elements.

Centaur Redesign for Mars Round Trip

Feb–Jun 2022

Redesigned a LOX/methane Centaur upper stage for a round-trip Mars mission, accounting for Martian in-situ resource utilisation.

BATES Motor Pressure Trace Analysis

May 2023

Analysed experimental solid-motor pressure traces to estimate maximum expected operating pressure and uncertainty for a 70-bar variant.

Impulsive Orbital-Transfer Trade Study

Oct 2021–Nov 2021

Analysed and compared multiple impulsive transfer strategies between assigned geocentric orbits using MATLAB, evaluating trade-offs in maneuver cost and transfer time.

RESEARCH INTERESTS

AI-enabled guidance, navigation and control; vision-based spacecraft navigation; uncertainty-aware state estimation and sensor fusion; rendezvous and proximity operations; small-body exploration; interpretable and physics-informed autonomy; adaptive and learning-enhanced control.

PROFESSIONAL STRENGTHS

Teamwork · Leadership · Quick Learning · Organisation · Drive · Quality Focus